

DEPARTMENT OF MECHANICAL ENGINEERING

B.Tech. Mech Engg IV SEMESTER Strength of Materials -II (PCC-ME-404-21) 2024-25

Time: 1.5 Hours

Max. Marks:15

Class test 1

Attempt all questions

R. 1. Drive compatibility equations for strain

5 CO2

Q.2. State of stress is given by following points.

$$\sigma_x = 80, \sigma_y = 100, \sigma_z = 120, \tau_{xy} = -60, \tau_{yz} = 40 \text{ and } \tau_{xz} = 20 \text{ MPa}$$

Determine the principal stresses. Also, check on their invariances.

5 CO2

Q3. Explain following theories of failure

5 CO4

- Rankine theory of failure,
- St. Venant theory of failure,
- Tresca's Theory of failure